

Incorrect

3. Question**1 point(s)**

In classful addressing, an IP address **123.23.156.4** belongs to _____ class format.

☐ C

☐ B

☒ A

☐ D

Correct

4. Question**1 point(s)**

What is the function of an Artificial Intelligence “Agent”?

☒ Mapping of environment sequence to an action

☐ Mapping of precept sequence to an action

☐ Work without the direct interference of the people

☐ Mapping of goal sequence to an action

Incorrect

5. Question**1 point(s)**

Write Your Name and IDNo.

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This response will be awarded full points automatically, but it can be reviewed and adjusted after submission.

Grading can be reviewed and adjusted.

6. Question**1 point(s)**

A repeater takes a weakened or corrupted signal and it.

☐ Resample

☐ Regenerates

☒ Amplifies

☐ Reroute

Incorrect

7. Question**1 point(s)**

-----are functions that are defined inside a class that describe the behaviors of an object.

☐ Object

☒ Method

☐ Class

☐ Attributes

Correct

8. Question

1 point(s)

In scheduling, the ready queue is maintained as a circular queue

☒ Priority Scheduling☐ RR☐ FCFS☐ SJF**Incorrect**

9. Question

1 point(s)

Caching is:

☐ copying information into faster storage system permanently☐ copying information into faster storage system temporarily☐ copying information from faster storage system permanently☒ copying information from slower storage system temporarily**Incorrect**

10. Question

1 point(s)

In LISP, the addition of 5+8 is entered as_____.

☐ 5 add 8☐ 5+8☒ (+5 8)☐ 5+8=**Correct**

11. Question**1 point(s)**

Repeater function in the layer.

- ☐ Physical
- ☒ Data link
- ☐ Network
- ☐ None of these

Incorrect**12. Question****1 point(s)**

In (student_id, name, address(street, city, state, zip-code)), address is

- ☒ a multivalued attribute.
- ☐ a relational attribute
- ☐ a derived attribute.
- ☐ a composite attribute.

Incorrect**13. Question****1 point(s)**

Using the same names with different parameters is known as:-

- ☐ Class
- ☐ Overriding
- ☒ Overloading
- ☐ Inheritance

Correct

14. Question**1 point(s)**

Which statement is true?

- ☐ None of above
- ☐ A relation can have different values at different times.
- ☐ A relation variable is inherently a specific set of values.
- ☒ A tuple corresponds to a row in a table.

Correct

15. Question**1 point(s)**

The design issue of Datalink Layer in OSI Reference Model is

- ☐ Representation of bits
- ☒ Framing
- ☐ Synchronization of bits
- ☐ Connection control

Correct

16. Question**1 point(s)**

Which statement is correct about do while statement?

- ☐ In do while no condition is required
- ☒ The condition is checked after the statement.
- ☐ The statement is checked before the statement.

- ☐ Do while is used to select program statements.

Correct

17. Question

1 point(s)

Which of the following best defines a class?

- ☒ Parent of an object

- ☐ Scope of an object

- ☐ Instance of an object

- ☐ Blueprint of an object

Incorrect

18. Question

1 point(s)

The worst case occurs in linear search algorithm when_____

- ☒ Item is the last element in the array or item is not there at all

- ☐ Item is the last element in the array

- ☐ Item is not in the array at all

- ☐ Item is somewhere in the middle of the array

Correct

19. Question

1 point(s)

The part of a database management system which ensures that the data remains in a consistent state is

- ☐ authorization and integrity manager

- ☐ buffer manager

☐ file manager☒ transaction manager**Correct****20. Question****1 point(s)**

Which of the given language is not commonly used for AI?

☐ PROLOG☐ LISP☒ Perl☐ Python**Correct****21. Question****1 point(s)**

Which one is the most likely advantage of UDP over TCP?

☐ Speed☐ Reliability☐ Sequenced☒ Acknowledgment**Incorrect****22. Question****1 point(s)**

Encapsulation and abstraction differ as _____.

☐ None of the above

☐ Hiding and Binding respectively

☒ Binding and Hiding respectively

☐ Hiding and hiding respectively

Correct

23. Question

1 point(s)

Which of the following is required to communicate between two computers?

☒ all of above including access to transmission medium

☐ communication hardware

☐ protocol

☐ communications software

Correct

24. Question

1 point(s)

_____ is when the system attempts to keep processes on the same processor but makes no guarantees.

☐ Soft affinity

☐ Push migration

☐ Pull migration

☒ Hard affinity

Incorrect

25. Question

1 point(s)

Match the following

- a. completeness i) how long does it take to find a solution
- b. Time complexity ii) how much memory need to perform the search
- c. Space complexity iii) is the strategy guaranteed to find the solution when there in one

☐ a-iii, b-ii ,c-i

☐ a-i, b-ii ,c-iii

☒ a-iii, b-i ,c-ii

☐ a-i, b-iii ,c-ii

Correct

26. Question

1 point(s)

Which of the following is true for network structure?

☒ It is a physical representation of the data.

☐ It allows many to many relationship.

☐ It is conceptually simple.

☐ It will be the dominant database of the future.

Correct

27. Question

1 point(s)

Which of the following is the main limitation for not running a Web server over a small home network connected to the Internet with a cable modem:

☒ Security risk

☐ Service agreement with ISP

☐ Software limitations.

☐ Hardware limitations.

Incorrect

28. Question**1 point(s)**

Which of the following is not correct?

- ☐ SQL data definitions may be used for defining primary keys as well as foreign keys
- ☐ SQL may be used for defining base tables as well as view tables
- ☐ SQL may be used for data definition as well as data retrieval
- ☒ SQL select operation may be used for data retrieval as well as for data modification.

Correct

29. Question**1 point(s)**

DDL is typically used during which phases of the development process?

- ☐ Physical design
- ☐ Logical Design
- ☒ Conceptual Design
- ☐ Implementation

Incorrect

30. Question**1 point(s)**

Source routing bridges in the same LANs must have bridge Number.

- ☐ Different
- ☒ Same
- ☐ Source

☐ Destination

Incorrect

Source routing bridges in the same LANs must have bridge Number. A. Same B. Different C. Source D. Destination Answer: B

31. Question

1 point(s)

Adaptive routing algorithms get their information from

☐ only from local environment

☐ only from external routers

☐ only from adjacent routers

☒ from locally, adjacent, external routers

Correct

32. Question

1 point(s)

The DNS name space is divided into non overlapping

☐ regions

☒ blocks

☐ zones

☐ divisions

Incorrect

33. Question

1 point(s)

Which of the following statements about a binary tree is correct?

☐ Every binary tree is either complete or full

☐ Every complete binary tree is also a full binary tree

☒ Every full binary tree is also a complete binary tree

☐ No binary tree is both complete and full

Correct

34. Question

1 point(s)

The operation of processing each element in the list is known as_____

☒ Traversal

☐ Inserting

☐ Merging

☐ Sorting

Correct

35. Question

1 point(s)

Which of the following constitutes a basic set of operations for manipulating relational

☐ SQL

☐ Predicate calculus

☒ Relational algebra

☐ Relational calculus

Correct

36. Question

1 point(s)

Which among the following is not a necessary condition for constructors?

☐ Its name must be same as that of class

☐ It must not have any return type

☐ It must contain a definition body

☒ It can contain arguments

Incorrect

37. Question

1 point(s)

The data or variables, defined within a **class** are called,

☐ Public

☐ overloading

☐ Package

☒ Instance variables

Correct

38. Question

1 point(s)

Computer Network is _____.

☐ Interconnected by communication channels

☐ Collection of hardware components and computers

☐ Sharing of resources and information

☒ All of the Above

Correct

39. Question

1 point(s)

The relationship between DEPARTMENT and EMPLOYEE is a

☒ one-to-many relationship

☐ one-to-one relationship

☐ many-to-one relationship

☐ many-to-many relationship

Correct

40. Question

1 point(s)

Which of the following syntax used to create an object of Class in Java?

☐ Class_name object_name = new() Class_name();

☐ Class_name object_name() = new Class_name()

☒ Class_name object_name = new Class_name();

☐ Class_name object_name = new Class_name;

Correct

41. Question

1 point(s)

Which of the following gives a logical structure of the database graphically?

☐ Entity diagram

☒ Database diagram

☐ Architectural representation

☐ Entity-relationship diagram

Incorrect

42. Question**1 point(s)**

A binary tree in which all its levels except the last, have maximum numbers of nodes, and all the nodes in the last level have only one child it will be its left child. Name the tree.

☒ Full binary tree☐ Complete binary tree☐ Threaded tree☐ M-way search tree**Incorrect****43. Question****1 point(s)**

Which one of the following is not relational operator?

☐ >=☐ ==☒ <<☐ None**Correct****44. Question****1 point(s)**

A parentheses checker program would be best implemented using _____

☐ Any of the above☐ List☐ Stack☒ Queue

Incorrect

45. Question**1 point(s)**

Which of the following is not a type of Artificial Intelligence agent?

☒ Simple reflex AI agent

☐ Learning AI agent

☐ Unity-based AI agent

☐ Goal-based AI agent

Incorrect

46. Question**1 point(s)**

_____ is the size difference in memory when we divide memory to fixed partitions and the allocated memory may be slightly larger than requested memory.

☒ External Fragmentation

☐ Internal Fragmentation

☐ Compaction

☐ Paging

Incorrect

47. Question**1 point(s)**

An entity set that does not have sufficient attributes to form a primary key is termed a_____

☐ Strong entity set

☐ Variant set

☒ Weak entity set

☐ Variable set

Correct

48. Question

1 point(s)

The method of network routing where every possible path between transmitting and receiving DTE is used is called

☐ Random Routing

☒ Packet Flooding

☐ Message Switching

☐ Directory Routing

Correct

49. Question

1 point(s)

What makes java platform independent?

☐ Class

☐ Object

☐ Bytecode

☒ simple

Incorrect

50. Question

1 point(s)

Minimum number of fields in each node of a doubly linked list is ___

☐ 4

☒ 3

☐ 0☐ 2**Correct****51. Question****1 point(s)**

From the following alternatives which programming language is highly portable?

☒ Java☐ Assembly language☐ C☐ C++**Correct****52. Question****1 point(s)**

In the drawing below of the Computer System Components, select the correct choice:

☒ 1. Users 2. Applications 3. OS 4. Hardware☐ 1. Users 2. OS 3. Applications 4. Hardware☐ 1. OS 2. Hardware 3. Applications 4. Users☐ 1. Hardwar 2. OS 3. Applications 4. Users

Incorrect

53. Question**1 point(s)**

It is an abstraction through which relationships are treated as higher level entities

☒ Specialization

☐ Generalization

☐ Inheritance

☐ Aggregation

Incorrect

54. Question**1 point(s)**

A graph is a tree if and only if graph is

☐ Directed graph

☐ Contains no cycles

☒ Planar

☐ Completely connected

Incorrect

55. Question**1 point(s)**

In Priority Scheduling, internal priorities are assigned by

☐ Users

☐ OS

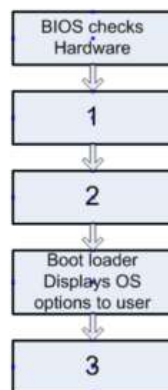
☒ System Programs

☐ CPU**Incorrect****56. Question****1 point(s)**

If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route is known as,

☐ Routing principle☐ Sink tree principle☒ Optimality principle☐ Network principle**Correct****57. Question****1 point(s)**

In the drawing of the Computer Startup Operation, select the correct choice:

☐ 1. MBR Loads BIOS 2. BIOS Loads boot loader 3. Boot loader Loads OS☐ 1. BIOS Loads OS 2. OS Loads boot loader 3. boot loader loads MBR☒ 1. BIOS Loads MBR 2. MBR Loads boot loader 3. Boot loader Loads OS

- ☐ 1. BIOS Loads boot loader 2. boot loader loads MBR 3. MBR Loads OS

Incorrect

58. Question

1 point(s)

What is an AI 'agent'?

- ☐ Takes input from the surroundings and uses its intelligence and performs the desired operations

- ☒ All of the mentioned

- ☐ An embedded program controlling line following robot

- ☐ Perceives its environment through sensors and acting upon that environment through actuators

Correct

59. Question

1 point(s)

What is Artificial Intelligence?

- ☒ Artificial Intelligence is a field that aims to develop intelligent machines

- ☐ Artificial Intelligence is a field that aims to improve the security

- ☐ Artificial Intelligence is a field that aims to make humans more intelligent

- ☐ Artificial Intelligence is a field that aims to mine the data

Correct

60. Question

1 point(s)

Every node **N** in a binary tree **T** except the root has a unique parent called the _____ of **N**.

- ☐ Antecedents

- ☒ Predecessor

☐ Forerunner☐ Precursor**Correct****61. Question****1 point(s)**

The default subnet mask of class **100.10.2.1** IP is

☒ 255.0.0.0☐ 255.255.0.0☐ 255.255.255.255☐ 255.255.255.0**Correct****62. Question****1 point(s)**

The one is a media access technique that identifies a collision and doing nothing

☐ Collision avoidance☐ Collision prevention☐ CSMA/CD☒ CSMA/CA**Incorrect****63. Question****1 point(s)**

What is a Firewall in Computer Network?

☐ An operating System of Computer Network

☐ The physical boundary of Network

☒ A system designed to prevent unauthorized access

☐ A web browsing Software

Correct

64. Question

1 point(s)

Logical address is the address generated by the _____.

☐ CPU

☐ MMU

☐ Cache Memory

☒ I/O

Incorrect

65. Question

1 point(s)

_____ implies a system can perform more than one task simultaneously on multi-core system

☐ Scalability

☐ Responsiveness

☒ Parallelism

☐ Concurrency

Correct

66. Question

1 point(s)

Which java variable declaration is **incorrect**?

☒ `int a=3.5;`☐ `float a=3,b=6.5;`☐ `int a,b,c;`☐ All**Correct****67. Question****1 point(s)**

How many objects can be declared in a specific class in a single program?

☒ As many as you want☐ 5☐ 2☐ 1**Correct****68. Question****1 point(s)**

System calls are:

☐ None of the above☒ Hardware interrupts☐ System Programs☐ Programming interface**Incorrect****69. Question****1 point(s)**

What is Weak AI?

- ☐ all of the mentioned
- ☒ a set of computer programs that produce output that would be considered to reflect intelligence if it were generated by humans
- ☐ the embodiment of human intellectual capabilities within a computer
- ☐ the study of mental faculties using mental models implemented on a computer

Incorrect

70. Question

1 point(s)

Hardware interrupts usually occur by:

- ☐ executing the last instruction in an application
- ☐ sending a signal to the CPU from specified device.
- ☐ starting a hardware device
- ☒ executing a system call from the application program

Incorrect

71. Question

1 point(s)

_____ is a more generalized single source shortest path algorithm which can find the shortest path in a graph with negative weighted edges.

- ☐ Bellman ford algorithm
- ☐ Dijkstra algorithm
- ☐ Prim's algorithm
- ☒ Kruskal's algorithm

Incorrect

72. Question

1 point(s)

In order traversing a tree resulted E A C K F H D B G; the preorder traversal would return

☐ FAEKDBHG☐ FEAKDCHBG☒ EAFKHDCBG☐ FAEKCDHGB**Incorrect**

73. Question

1 point(s)

Military vehicles on a battlefield with no existing infrastructure will deploy network.

☐ LAN☐ Cell Network☐ WAN☒ MANET**Correct**

74. Question

1 point(s)

Which statement is false?

☐ Functional dependencies (FDs) are used to specify formal measures of the "goodness" of relational designs.☒ If the attribute X determines the attribute Y, X is functionally dependent on Y.☐ Functional dependencies and keys are used to define normal forms for relations.

- ☐ Functional dependencies are constraints that are derived from the meaning and interrelationships of the data attributes

Correct

75. Question

1 point(s)

Why do we need to handle exceptions?

- ☒ To prevent abnormal termination of program

- ☐ To encourage exception prone program

- ☐ To avoid syntax errors

- ☐ To save memory

Correct

76. Question

1 point(s)

In transmission, the channel capacity is shared by both communicating devices at all times.

- ☐ automatic

- ☐ Simplex

- ☐ half-duplex

- ☒ full-duplex

Correct

77. Question

1 point(s)

Which of these media gives the highest bit rates over the longest distances?

- ☐ Unshielded twisted pair cable

- ☐ Shielded Twisted pair cable

☐ Coaxial cable☒ Optical fiber cable**Correct****78. Question****1 point(s)**

Which point below is not a way for handling Deadlocks?

☒ Transfer☐ Ignore☐ Prevention☐ Detection and Recovery**Correct****79. Question****1 point(s)**

Which of the following can improve the performance of an AI agent?

☒ All of the mentioned☐ Observing☐ Learning☐ Perceiving**Incorrect****80. Question****1 point(s)**

If one variable is declared by using private access modifier, this variable can be accessed in:

☐ The same class

☐ Different packages☐ Sub-class☒ The same packages**Incorrect****81. Question****1 point(s)**

In crossover cable which wires are crossed in side A and side B

☒ 1236☐ 1357☐ 2468☐ 1234**Correct****82. Question****1 point(s)**

One of the following is the limitation of object-oriented programming?

☒ Not suitable for small problems☐ All☐ Takes more time to solve problems☐ Solving problems takes more time☐ Requires intensive testing processes**Incorrect****83. Question****1 point(s)**

Which of the following is an external sorting?

☒ Tree sort

☐ Merge sort

☐ Insertion sort

☐ Bubble sort

Incorrect

84. Question

1 point(s)

Which of the following remote access programs is the most secure:

☐ rlogin

☐ telnet

☐ ssh

☒ ftp

Incorrect

85. Question

1 point(s)

Which of the following is not an application of artificial intelligence?

☒ DBMS

☐ LIDAR

☐ Face recognition system

☐ Chatbots

Correct

86. Question

1 point(s)

A graph in which all vertices have equal degree is known as ____

- ☐ Simple graph
- ☒ Complete graph
- ☐ Multi graph
- ☐ Regular graph

Correct

87. Question

1 point(s)

The advantage of a client /server network over a peer to peer network is

- ☐ Performances is better on peer to peer
- ☒ Security is better on client/server
- ☐ Easy to configuration the client/ server
- ☐ Security is better on peer to peer

Correct

88. Question

1 point(s)

The time complexity of quicksort is _____

- ☒ $O(n \log n)$
- ☐ $O(n)$
- ☐ $O(n^2)$
- ☐ $O(\log n)$

Correct

89. Question

1 point(s)

Does index take space in the disk?

☐ it stores memory as and when required

☐ indexes are never stored on disk

☐ indexes take no space

☒ yes, indexes are stored on disk

Correct

90. Question

1 point(s)

In SQL the statement **select * from R, S** is equivalent to

☐ Select * from R cross join S

☐ Select * from R union join S

☒ Select * from R natural join S

☐ Select * from R inner join S

Incorrect

91. Question

1 point(s)

What is data integrity?

☐ It is the data contained in database that is secured.

☒ It is the data contained in database that is shared.

☐ It is the data contained in database that is accurate and consistent.

☐ It is the data contained in database that is non redundant.

Incorrect

92. Question**1 point(s)**

The clause alter table in SQL can be used to

- ☐ alter the default values of an attribute
- ☐ delete an attribute
- ☒ all of the above
- ☐ add an attribute

Correct

93. Question**1 point(s)**

What problems could information is stored redundantly in a database?

- ☐ Deletion anomalies
- ☐ Modification anomalies
- ☒ All of the above
- ☐ Insertion anomalies

Correct

94. Question**1 point(s)**

The correct order of data decapsulation is

- ☐ segment→packet→data→bits→frame
- ☒ Bit→frame→packet→segment→data
- ☐ Data→segment→packet→frame→bit

☐ Frame→bit→data→segment→packet

Correct

95. Question

1 point(s)

In $(\text{student_id}, \text{name}, \text{address}(\text{street}, \text{city}, \text{state}, \text{zip-code}))$, address is

☐ a derived attribute.

☒ a multivalued attribute.

☐ a composite attribute.

☐ a relational attribute

Incorrect

96. Question

1 point(s)

Placing multiple processors on same physical chip, is called .

☐ Multitaskin

☐ Multiprogramming

☐ Multicore

☒ Asymmetric multiprocessing

Correct

97. Question

1 point(s)

In a circular queue the value of rear (r) will be

☒ $r = r+1$

☐ $r = (r+1) \% [\text{QUEUE_SIZE} - 1]$

☐ $r = (r+1) \% \text{QUEUE_SIZE}$ ☐ $r = (r-1) \% \text{QUEUE_SIZE}$

Incorrect

98. Question**1 point(s)**

The situation when in a linked list $\text{START} = \text{NULL}$ is

☐ Overflow☒ Underflow☐ Houseful☐ Saturated

Correct

99. Question**1 point(s)**

An operating system is a program that:

☐ It provides an environment within which computer hardware can do work☒ manages the computer hardware and provides a basis for application programs☐ manages the application programs and provides a basis for computer hardware☐ manages both application programs and computer hardware

Incorrect

100. Question**1 point(s)**

In select student_id, score from the grade_report table where course_code = 'CoSc2041', what is course_code = ' CoSc2041'?

☐ None of the above

☐ A condition☒ A list of attribute names☐ A list of the relation names**Incorrect****101. Question****1 point(s)**

Which feature of Object-Oriented Programming indicates code reusability?

☐ Encapsulation☐ Polymorphism☐ Abstraction☒ Inheritance**Correct**